Airbnb Vacation Rental Architecture

Simple, Scalable Production Blueprint (Clean 4-Tier Flow)

1. CLIENTS

- Desktop Web Browser | Mobile Web & Apps | React 18 + Tailwind CSS + Framer Motion

2. GLOBAL CDN & ASSET CACHE

- Cloudflare Edge PoPs: Sub-10ms global latency
- Responsive image optimization & WebP format conversion

3. API GATEWAY

- User authentication & token validation
- Rate limiting, request routing & security checks

Listing Service

- Property specs & rooms
- 5-Photo bento hero
- Amenities & host rules

Booking & Availability

- Real-time calendar engine
- 10-sec checkout lock
- Zero double-booking

Search & Location

- Multi-city Goa search
- Map coordinates & pins
- Guest & price filters

Pricing & Reviews

- Nightly rates + clean fee
- Tax calculations
- 6-category review bars

PostgreSQL Database

- ACID transactional store for reservations
- Listing details, verified guests & reviews
- Multi-region read replicas

Redis Cache & Locks

- Distributed lock (Redlock) during checkout
- Fast calendar availability lookups
- Listing metadata caching

OpenSearch & Kafka

- Instant map search & radius discovery
- Kafka real-time event pipeline
- Zero-delay inventory synchronization